

Songhang Li

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New York, NY

EDUCATION

Columbia University <i>B.S. in Electrical Engineering</i>	New York, NY May 2027
Franklin & Marshall College <i>B.A. in Physics and Mathematics Magna Cum Laude ΦBK</i>	Lancaster, PA May 2025
University of California Berkeley <i>Visiting, Physics</i>	Berkeley, CA Summer 2023

AWARDS & HONORS

Frank Durrell Enck Memorial Prize in Physics <i>F&M</i> Established in 1975; traditionally awarded to a graduating Physics major whose overall departmental performance and involvement is most deserving of recognition.	Fall 2024
Kershner Scholar in Mathematics <i>F&M</i> Awarded to students based on their proficiency in Mathematics.	Spring 2025
Phi Beta Kappa (ΦBK) <i>U.S.A.</i> Invitation-only academic honor society recognizing excellence in the liberal arts and sciences.	Since 2025
The Nolt Music Award <i>F&M</i> Awarded to F&M student musicians to undertake musically enriching projects such as a senior recital.	Spring 2025
The Alma Mater Singer <i>F&M (238th Commencement)</i> Selected as the Alma Mater Singer for Franklin & Marshall College's 238th Commencement.	Spring 2025
Summer Scholar Award <i>F&M</i> Awarded to a student researcher for summer research conducted with faculty.	Summer 2024
Pi Mu Epsilon (Math Honor Society) <i>U.S.A.</i> Nationally recognized honorary society dedicated to furthering knowledge of mathematics.	Since Spring 2024
Sigma Pi Sigma (Physics & Astronomy Honor Society) <i>U.S.A.</i> Honors outstanding scholarship in physics and astronomy.	Since Spring 2023
Student Independent Research Grants <i>F&M</i> Awarded for advanced student research/projects whose needs exceed departmental budgets.	Fall 2023
Kershner Scholar in Physics <i>F&M</i> Awarded to students based on their proficiency in Physics.	Spring 2023
Dean's and Honors Lists <i>F&M</i> Earned Honors List recognition for achieving a 3.7 or better grade point average.	Since Fall 2021

PUBLICATIONS

J.K. Krebs, **Songhang Li**, Dragos Georgescu, and T. D. Pham. (*preprint, under review*). *Eu3+ Luminescence as a Probe of Local Crystallinity in Combustion-Synthesized LaAlO₃*. *SSRN*, [doi:10.2139/ssrn.6102568](https://doi.org/10.2139/ssrn.6102568)

Yuxuan Du, Wenya Du, Dabin Lin, Minghao Ai, **Songhang Li**, Lin Zhang. (2023). *Recent Progress on Hydrogel-Based Piezoelectric Devices for Biomedical Applications*. *Micromachines*, 14(1). [doi:10.3390/mi14010167](https://doi.org/10.3390/mi14010167)

PRESENTATIONS & CONFERENCES

- Mid-Atlantic Section of the American Physical Society Meeting** | *Temple University, PA* Nov. 15, 2024
- Rate equation model for fluorescence and phosphorescence in Eu^{3+} doped LaAlO_3 .
- Student Research Fair** | *Franklin and Marshall College, PA* Oct. 18, 2024
- Photon Counting Experiment of Persistent Luminescence in Eu:LAP
- Student Research Fair** | *Franklin and Marshall College, PA* Apr. 20, 2024
- Design, construction, testing, and improving: An introduction to Engineering Design Competitions.

ACADEMIC & RESEARCH EXPERIENCES

- Photon Counting Experiment of Persistent Luminescence in Eu:LAP (prep.)** May 2024 – Jan. 2025
Research Assistant Advisor: Prof. Ken Krebs *F&M, Lancaster, PA*
- Measured decay dynamics of excited population of impurity ions after UV laser excitation using photon counting.
 - Developed a circuit solution using Arduino to precisely control excitation laser mechanical shutter, enabling photon counting of early state decay.
 - Compared the emission spectra measurement with calculated energy levels of Eu^{3+} in LaAlO_3 using density functional theory.
 - Conducted a literature review and originated a table summarizing experimental technique and materials.
- Society of Automotive Engineers Aero Design Competition Preparation** Sept. 2023 – May 2025
Team Leader, Pilot Advisor: Prof. Ken Krebs *F&M, Lancaster, PA*
- Designed a power and lift efficient, short-take-off-length 3.8 meters fixed wing plane for water delivery.
 - Calculated the lift of the wing based on the structure of air foil in order to improve the load capacity.
 - Fabricated the wings utilizing balsa wood and carbon fiber, employing laser cutter and heat-shrink covering film application techniques.
 - Led the team members to write the design report and presented the result in a research fair.
 - Wrote a funding proposal and managed to receive a year-long student research grant from the grant committee.
- Automation Algorithm for 4 DOF Robotic Arm** Jan. 2023 – May 2023
Researcher Advisor: Prof. Ken Krebs *F&M, Lancaster, PA*
- Searched recent publications, textbooks, GitHub robotic arm fabrication and algorithm resources to assist designing a research outline for the group project.
 - Designed and built a circuit for touch sensing of a grasping mechanism utilizing piezoresistance sensors.
- Electrical and Optical Measurement of Ferroelectrics Material (PLZT)** Sept. 2022 – Dec. 2022
Researcher Advisor: Prof. Ken Krebs *F&M, Lancaster, PA*
- Investigated the relation between the refractive index and dielectric constants of PLZT film.
 - Built multilayer models by ellipsometry data analysis software CompleteEASE to model measurements of the thickness of thin films.
 - Measured the refractive index using J.A. Woollam M-2000 spectrum ellipsometry.
 - Made a two-plate capacitor with PLZT as dielectric layer to obtain the dielectric constant from capacitance measurements.
 - Compared optical and electrical techniques for determination of dielectric constant and refractive index.
- Universal Village Program-Smart Materials and Intelligent Electronics** Jun. 2022 – Aug. 2022
Research Assistant Advisor: Dr. Lin Zhang, Massachusetts Institute of Technology Media Lab *Boston, MA*
- Designed a flexible electrode pattern exposure mask using AutoCAD, in preparation for printing of mask.
 - Prepared piezoelectric 1-3 composites for piezoelectric sensors.
 - Organized the piezoelectric hydrogel sensors comparison table and contributed to manuscript writing, see publication (*Micromachines* 2023, 14, 167).

ACTIVITIES

- Vocal** Sept. 2021 – May 2025
Recital & College Chorus Singer *F&M*
- Sang the Alma Mater at the Class of 2025 Commencement.
 - Performed with the College Chorus at major campus events (e.g., True Blue Weekend).
 - Performed in end-of-semester vocal lesson recitals each semester.
- Alumni Leaders of Tomorrow, Office of Alumni Engagement** Jun. 2023 – May 2024
Member *F&M*
- Created alumni connection and networking programs to support student–alumni engagement.
- F&M Tennis Club** Sept. 2023 – May 2025
Member *F&M*
- Trained and onboarded new club members in collaboration with the coach.

SKILLS & INSTRUMENTATION

Hardware & Instrumentation: Oscilloscopes, Waveform Generators, FPGA, HP LCR, M-2000 J.A. Woollam Spectrum ellipsometry, Monochromator SR430, Stanford Research System Multichannel Photon Counter.

EDA Tools & Design Software: SPICE simulation, MATLAB, Simulink, Arduino IDE, AutoCAD, Autodesk Fusion

Programming language: Python, Java, VHDL, MATLAB, MIPS Assembly, Mathematica, LaTeX, Arduino, R.

Software: SPICE, IntelliJ, AutoCAD, MATLAB, CompleteEASE(optical modeling and ellipsometry data fitting)

Design & Analysis Skills: Analog and digital circuit simulation (AC/DC/transient analysis), Circuit design and debugging, PCB layout and soldering, signal characterization, Data analysis with Python/MATLAB/SciDavis, Characterization of electronic components and semiconductor materials, Embedded prototyping and validation,

Additional Technical Skills: Homemade PCB including exposure and etching technics, Laser cutting/fabrication (Glowforge), technical documentation (LaTeX, Endnote), design visualization and project planning (X-mind), image processing (Adobe Lightroom, Photoshop)

Language: Mandarin native, Hakka native.

EXPERIENCE

- Department of Electrical Engineering** Feb. 2026 – Present
Course Assistant *Columbia University, New York, NY*
- Preparing the LED matrix project using Arduino for ENGI E1102: *The Art of Engineering*.
- Department of Physics and Astronomy** Jan. 2024 – May 2025
Teaching and Research Assistant *Franklin & Marshall College, Lancaster, PA*
- Compared two electronics textbooks and exercises to identify key instructional concepts for PHY321 Intro to Electronics.
 - Tested different teaching kits selected by the professor and evaluated suitability for effective teaching and learning.
- Career Center** Sept. 2022 – May 2025
Photographer *Franklin & Marshall College, Lancaster, PA*
- Captured LinkedIn headshots and event photography for Career Services programming.

INTEREST

RC airplane, singing, photography (fashion and documentary, film, and darkroom processing), tennis, swimming, cycling.